

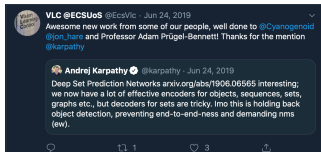
**Make a  
forward pass  
before the  
backward pass**

# Backpropagation: Understanding the implications of the chain rule

Jonathon Hare

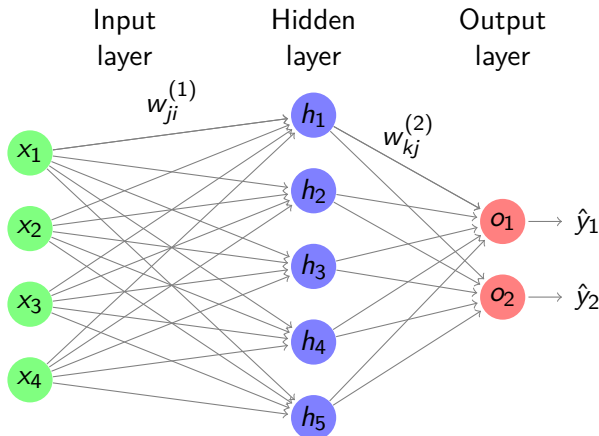
Vision, Learning and Control  
University of Southampton

A lot of the ideas in this lecture come from Andrej Karpathy's blog post on backprop (<https://medium.com/@karpathy/yes-you-should-understand-backprop-e2f06eab496b>) and his CS231n Lecture Notes (<http://cs231n.github.io/optimization-2/>)



- A quick look at an MLP again
- The chain rule (again)
- Unintuitive gradient effects
- A closer look at basic stochastic gradient descent algorithms

# The bias-term free Multilayer Perceptron



Without loss of generality, we can write the above as:

$$\hat{\mathbf{y}} = g(f(\mathbf{x}; \mathbf{W}^{(1)}); \mathbf{W}^{(2)}) = g(\mathbf{W}^{(2)} f(\mathbf{W}^{(1)} \mathbf{x}))$$

where  $f$  and  $g$  are activation functions.

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$$\ell_{MSE}(\hat{\mathbf{y}}, \mathbf{y}) = \|\hat{\mathbf{y}} - \mathbf{y}\|_2^2$$

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- We could do this by hand
- We're not that crazy! We've already seen how AD reduces this to a problem of computing JVPs (or better VJPs here - the reverse mode)
- But, what does that mean in terms of an algorithm for computing the gradients?

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$$\text{so } \nabla_{[x, y, z]} f = [z, z, q]$$

# A computational graph perspective

$$f(x, y, z) = (x + y)z$$

# An intuition of the chain rule and reverse mode AD

- Notice how every operation in the computational graph given its inputs can immediately compute two things:
  - ① its output value
  - ② the *local* gradient of its inputs with respect to its output value
- The chain rule tells us literally that each operation should take its local gradients and multiply them by the gradient that *flows* backwards into it

# This is backpropagation

- The backprop algorithm is just the idea that you can perform the forward pass (computing and caching the local gradients as you go),
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- Backprop is just another name for 'Reverse Mode Automatic Differentiation'...

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  - **Hence you need to always pay attention to data normalisation!**

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    - What's the implication of this in a deep network with sigmoid activations?

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- These are dead ReLUs - ones that never fire for all training data
  - Sometimes you can find that you have a large fraction of these
  - if you get them from the beginning, check weight initialisation and data normalisation
  - if they're appearing during training, maybe  $\lambda$  is too big?

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- Same thing happens in the backward pass of an RNN (although with matrices rather than scalars, so the reasoning applies to the largest eigenvalue)